

■ Energy-Aware Adaptive Multi-Objective WSN Coverage Optimization

Algorithm Formulation: EA-VVF-MOPSO & Downstream Hybrid LEACH-PEGASIS Routing

Liquid Glass UI Master Reference Manual • Plain English & Full Algorithm Guide

■ **Executive Summary:** This project develops an intelligent software optimizer (**EA-VVF-MOPSO**) that eliminates redundant sensor overlaps and coverage blind spots in Wireless Sensor Networks while preventing battery depletion from excessive physical movement. Optimized nodes then communicate via a hybrid clustered-chain protocol (**Hybrid LEACH-PEGASIS**).

1. Master Glossary of Algorithm Full Forms

Acronym	Full Expanded Name	Role in Project & Intuitive Function
EA-VVF-MOPSO	Energy-Aware Adaptive Voronoi-Virtual Force Multi-Objective Particle Swarm Optimization	Main Proposed Method: Integrates global swarm search, Voronoi geometry, push/pull virtual magnets, and battery protection.
MOPSO	Multi-Objective Particle Swarm Optimization	Global Swarm Explorer: Balances competing objectives (high coverage vs. low overlap) using an external Pareto archive.
PSO	Particle Swarm Optimization	Single-Objective Baseline: Standard bird-flock swarm optimizing only a single scalar coverage value.
VFA	Virtual Force Algorithm	Micro-Adjuster: Physics-inspired forces that push overlapping nodes apart and pull nodes into blind spots.
Voronoi	Voronoi Diagram / Geometric Tessellation	Territory Radar: Slices the map into polygons to geometrically identify crowded clusters and empty holes.
LEACH	Low-Energy Adaptive Clustering Hierarchy	Clustering Protocol: Organizes sensors into local groups and rotates Cluster Heads to balance battery drain.
PEGASIS	Power-Efficient GATHERing in Sensor Information Systems	Chain Protocol: Passes data in a near-neighbor daisy chain to minimize radio transmission power.
GA	Genetic Algorithm	Evolutionary Baseline: Uses crossover and mutation on deployment chromosomes.
WSN	Wireless Sensor Network	A network of spatially dispersed, battery-powered sensor nodes communicating wirelessly.

2. Everyday Intuition: The Park Security Camera Analogy

The Problem: Imagine dropping 50 battery-powered security cameras randomly into a 100-acre public park. Because the drop is random, several cameras will land huddled together pointing at the exact same bench (**wasted redundant overlap**), while other areas like the playground or parking lot will have no camera at all (**coverage holes / blind spots**). If we ask cameras to drive across the park to fix this, their small batteries will run out before surveillance even starts.

The Solution: We use **EA-VVF-MOPSO** to compute the perfect positions on a computer first. The swarm finds the best layout, Voronoi maps detect blind spots, virtual magnets push overlapping cameras apart, and weak cameras are kept still. Once placed, cameras send data to the main security booth using **Hybrid LEACH-PEGASIS** (team leaders + relay chains) to maximize battery life.

3. Abstract Decoded: Academic Jargon vs. Plain English

Why does the academic abstract sound so complex? Here is a direct line-by-line translation into simple everyday English:

Academic Sentence (Jargon)	What It Actually Means (Plain English)
"Random deployment yields non-uniform spatial topologies, causing dense clusters of redundant overlap alongside unmonitored coverage holes."	When you scatter sensors randomly, some land in a pile watching the exact same spot (waste), while other areas have blind spots with no sensors.
"Conventional repositioning techniques optimize coverage in isolation, disregarding the physical energy expenditure of sensor displacement."	Other scientists try to make sensors roll/move to new spots, but moving takes huge battery power, draining sensors before data collection even starts.
"A global MOPSO layer searches non-dominated deployment topologies using an external Pareto archive..."	A simulated "flock of birds" tests hundreds of map layouts on a computer to find the best compromise (high coverage + low overlap + low battery drain).
"Voronoi tessellation pinpoints under-monitored regions, while Virtual Force applies attractive and repulsive adjustments under residual energy constraints."	We slice the map into territories to spot blind spots. Virtual magnets push overlapping nodes apart and pull nodes into empty spots. Low-battery nodes are locked in place.
"The network executes a Hybrid LEACH-PEGASIS protocol for energy-efficient data gathering."	Once in position, sensors group into teams and pass data along short relay chains to the Base Station so their batteries last as long as possible.
"Evaluated across 30+ seeds with Wilcoxon and Friedman non-parametric statistical tests."	We tested 30 different random setups to prove mathematically that our method consistently beats older algorithms and wasn't just lucky.

4. The 30-Second Viva Pitch (What to say to your Professor)

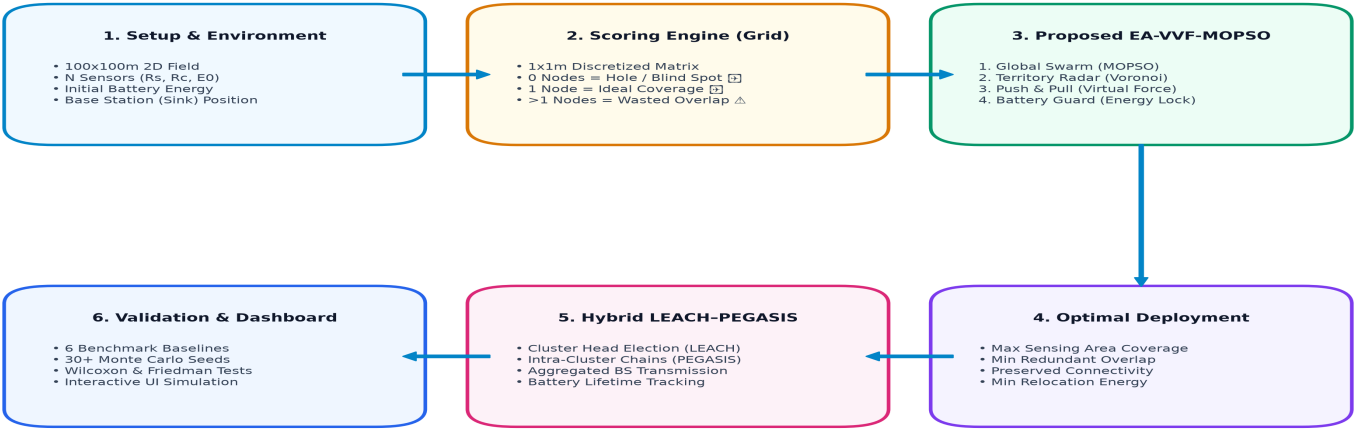
■■ Memorize this 3-sentence summary for your project review / viva:

"When sensors are deployed randomly in an area, they create wasted overlaps in some spots and blind spots in others. Moving them around blindly drains their batteries.

Our project builds **EA-VVF-MOPSO**, a smart computer algorithm that uses swarm intelligence, territory mapping, and virtual push/pull forces to find the perfect positions for sensors while protecting low-battery nodes.

Once positioned, the sensors send their data through an ultra-efficient **Hybrid LEACH-PEGASIS** cluster-chain protocol, giving the network maximum coverage and the longest possible battery life."

5. End-to-End System Block Diagram

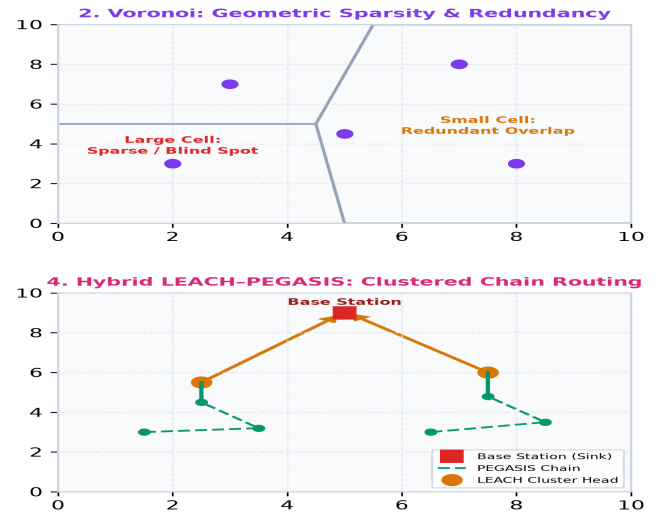
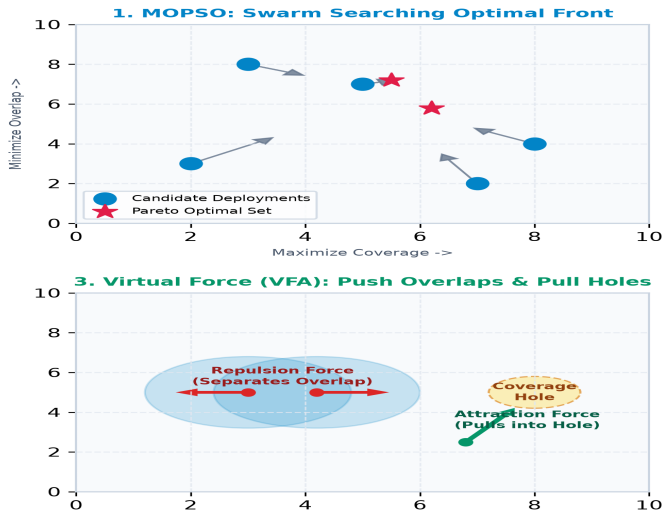


6. Step-by-Step Implementation Roadmap

The project is built across 6 self-contained, modular engineering steps:

Step & Phase	Technical Details & Implementation Tasks	Primary Deliverable
Step 1: Simulation Foundation	• Model Sensor class (x, y, R_s, R_c, E_0, E_{res}). • Create 2D Field matrix (100x100 m, 50 nodes). • Implement reproducible random seed generation.	Initial random deployment matrix
Step 2: Scoring Engine	• Discretize field into 1x1 m grid matrix. • Vectorized point count: 0 = Hole ■, 1 = Unique ■, >1 = Overlap ■■. • Calculate Coverage Ratio, Overlap Ratio, and Hole Ratio.	Coverage/Overlap heatmaps and scoring logs
Step 3: Baseline Algorithms	• Implement standard benchmarks: Random, Standard PSO, Genetic Algorithm (GA), and Virtual Force (VFA). • Run all under identical random seeds for fair comparison.	Benchmark comparison datasets
Step 4: Proposed EA-VVF-MOPSO	• Global MOPSO: Swarm search maintaining Pareto archive. • Voronoi: Geometric polygon tiling to detect density & holes. • Virtual Force: Attractive force to holes + Repulsive to overlaps. • Energy Guard: Locks displacement if $E_{res} < 30\%$.	Optimal node coordinates with high CR & low OR
Step 5: Hybrid LEACH-PEGASIS	• Cluster nodes dynamically via LEACH probability. • Construct intra-cluster PEGASIS daisy chains. • Aggregate data packets and transmit to Base Station. • Track FND, HND, LND, and Packet Delivery Ratio.	Network Lifetime curves (Alive Nodes vs. Rounds)
Step 6: Statistical Tests & UI	• Run 30+ Monte Carlo seeds across 5 density scenarios (25 to 150 nodes). • Wilcoxon Signed-Rank and Friedman significance tests. • High-resolution publication plots & live interactive dashboard.	Publication figures, CSV/JSON logs & Live GUI

7. Visual Guide to the Core Algorithmic Mechanics



8. Deep-Dive: The 4 Pillars of EA-VVF-MOPSO

A. Multi-Objective Particle Swarm Optimization (MOPSO) — The Global Swarm Explorer

- **Bird Flock Analogy:** Birds searching a mountain range for food; when one spots a great feeding zone, the flock shifts velocity toward that area.
- **Mechanism:** Each particle is a complete candidate network map $X = [x_1, y_1, x_2, y_2, \dots, x_N, y_N]$. MOPSO evolves a **Pareto Archive** of non-dominated layouts balancing coverage, overlap, holes, and movement cost simultaneously.
- **Velocity Update:** $v_i(t+1) = w \cdot v_i(t) + c_1 \cdot r_1 \cdot (pbest_i - x_i) + c_2 \cdot r_2 \cdot (gbest - x_i)$.

B. Voronoi Tessellation — The Territory & Crowding Radar

- **Coffee Shop Analogy:** Borders drawn so every citizen visits their closest shop. Huge cells = cold coffee (blind spots); tiny cells = overcrowding (overlap).
- **Mechanism:** Slices the continuous 2D field into closest-neighbor polygons. If $Area(Cell_i) > \pi R_s^2$, a coverage hole exists; if $Area(Cell_i) < \pi R_s^2$, node S_i is redundant and flagged for relocation.

C. Virtual Force Algorithm (VFA) — The Push & Pull Micro-Refiner

- **Floating Magnets Analogy:** Magnetic spheres repel like poles and are pulled toward vacuum zones.
- **Mechanism:** Computes force vector $F_{total} = F_{attraction} + F_{repulsion} + F_{boundary}$. Repulsion pushes overlapping nodes apart ($dist < 2 R_s$); attraction pulls nodes toward hole centroids; boundary forces keep nodes inside the field.

D. Energy-Aware Controller — The Battery Guard

- **Tired Player Analogy:** The coach keeps a player with 10% stamina stationary in defense rather than sprinting.
- **Mechanism:** Physical relocation consumes significant battery. If $E_{res} / E_{init} < 30\%$, the Energy Guard locks the node's coordinates.

9. Downstream Routing: Hybrid LEACH-PEGASIS Protocol

- **Full Names:** LEACH = *Low-Energy Adaptive Clustering Hierarchy*; PEGASIS = *Power-Efficient GATHERing in Sensor Information Systems*.
- **Classroom Exam Analogy:** 50 students submitting exams to the professor on stage. Direct individual walking (direct transmission) exhausts everyone. LEACH elects Section Leaders (Cluster Heads). PEGASIS forms single-file relay lines passing papers neighbor-to-neighbor using minimal energy. Only the Leader collects the combined papers and delivers them to the professor.
- **First-Order Radio Energy Model:**
 - Transmitting I bits over distance d : $E_{Tx}(I, d) = I \cdot E_{elec} + I \cdot \epsilon_{fs} \cdot d^2$ (free space) or $I \cdot E_{elec} + I \cdot \epsilon_{mp} \cdot d^4$ (multipath).
 - Receiving I bits: $E_{Rx}(I) = I \cdot E_{elec}$; Aggregation Energy: $E_{DA} = I \cdot E_{agg}$.
- **Hybrid Synergy:** Prevents energy-draining long-distance individual transmissions, significantly extending First Node Death (FND) and Half Node Death (HND).

10. Baseline Comparison Algorithms (Why Our Method Wins)

Baseline Algorithm	How It Operates	Why Our Method Beats It
1. Random Deployment	Sensors remain static at their initial drop coordinates without any relocation.	High blind spots, heavy overlap clusters, and rapid battery death.
2. Standard PSO (Particle Swarm Opt.)	Optimizes a single scalar coverage objective.	Ignores redundant overlap and causes excessive sensor travel battery drain.
3. Genetic Algorithm (GA)	Treats deployment as DNA chromosomes with crossover/mutation.	Slow convergence speed; struggles with fine continuous spatial adjustments.
4. VFA Alone	Relies solely on local push/pull forces between neighbor nodes.	Lacks global exploration; easily gets trapped in local deadlocks (local minima).
5. Voronoi Placement Alone	Moves sensors directly to centroids of Voronoi polygons.	Disregards sensing radius overlap and remaining battery levels.
6. Standard MOPSO	Multi-objective particle swarm without geometric/force refinement.	Slower spatial packing convergence without Voronoi/VFA cues.

11. Multi-Objective Mathematical Formulation

- **Objective 1 (Maximize Coverage Ratio):** $CR = N_{covered} / N_{total} \times 100$ (percentage of monitored 1×1m grid points).
- **Objective 2 (Minimize Overlap Ratio):** $OR = N_{redundant} / N_{covered}$ (fraction of covered points monitored by ≥2 sensors).
- **Objective 3 (Minimize Hole Ratio):** $HR = N_{uncovered} / N_{total} \times 100$.
- **Objective 4 (Maximize Residual Energy):** $E_{factor} = (1/N) \sum (E_{residual_i} / E_{initial_i})$.
- **Objective 5 (Maximize Network Connectivity):** $K = |Largest\ Connected\ Component| / N$.
- **Objective 6 (Minimize Sensor Movement Cost):** $M = \sum ||p_{i_final} - p_{i_initial}||$.
- **Composite Fitness:** $Fitness = w_C \cdot C - w_O \cdot O - w_H \cdot H + w_E \cdot E + w_K \cdot K - w_M \cdot M$ (tracked alongside non-dominated Pareto archiving).

12. Experimental Evaluation Matrix & Statistical Rigor

- **Network Density Scenarios:** S1 (50×50 m, 25 nodes), S2 (100×100 m, 50 nodes), S3 (100×100 m, 100 nodes), S4 (200×200 m, 100 nodes), S5 (100×100 m, 150 nodes).
- **30+ Monte Carlo Seeds:** Eliminates random deployment bias; all algorithms evaluated under identical initial node coordinates.
- **Statistical Tests:** **Wilcoxon Signed-Rank Test** (paired non-parametric comparison, α=0.05) and **Friedman Test** (multi-algorithm rank test).
- **Routing Metrics:** First Node Death (FND), Half Node Death (HND), Last Node Death (LND), Packet Delivery Ratio (PDR), and Network Throughput.

13. Expected Deliverables & Directory Structure

Category	Directory & File Contents
Source Engine	src/models/ (sensor.py), src/coverage/ (grid.py, voronoi.py), src/optimization/ (mopso.py, vfa.py, ea_vvf_mopso.py), src/routing/ (hybrid_leach_pegasis.py)
Telemetry Data	results/raw/ (per-seed CSV/JSON logs), results/metrics/ (statistical test summaries, FND/HND tables)
Figures & Dashboard	results/plots/ (deployment maps, coverage/overlap heatmaps, alive-node decay curves), app.py (interactive simulation dashboard)